Triangle constraints at two compact arithmetic rank–21 vertex types

Research manuscript with reproducible exact certificates

18 October 2026

Abstract

We prove local incidence restrictions at compact arithmetic hyperbolic Coxeter vertices with ground field $\mathbb{Q}(\sqrt{5})$ and diagram $\mathrm{A}_1+\mathrm{D}_5+\mathrm{E}_7+\mathrm{E}_8$ or $\mathrm{D}_5+2\mathrm{E}_8$. The first type has at most seven triangular two-faces marked by nonadjacent roots within one nontrivial component. At either type, at most one of the three possible nonadjacent D_5 triangles occurs. We describe forced simplex and product faces in the exceptional components. Exact finite-corner and arithmetic computations are supplied. These results do not improve a dimension ceiling or establish a new global charging inequality.

1 Statements and hypotheses

Roots have square two. Number D_5 by the edges $01, 12, 23, 24$. Number E_n by the path $0-1-\cdots-(n-2)$ and the extra edge $2(n-1)$. All these bonds have label three. A pair of incident facet roots specifies a two-face at a simple vertex. Call the pair nonadjacent when it is not an edge of its old component diagram. We consider only pairs in a single component.

Theorem 1. *Let $P \subset \mathbb{H}^{21}$ be a compact arithmetic Coxeter polytope with ground field $\mathbb{Q}(\sqrt{5})$, and let a vertex have diagram $\mathrm{A}_1+\mathrm{D}_5+\mathrm{E}_7+\mathrm{E}_8$. Among its 42 two-faces marked by nonadjacent pairs within $\mathrm{D}_5, \mathrm{E}_7, \mathrm{E}_8$, at most seven are triangles. Equality in the necessary local domain requires the E_7 third wall with label-five contact at root 5 and the two E_8 third walls with label-five contacts at roots 0 and 6. It has no nonadjacent D_5 triangle. Realization of this necessary equality configuration is not asserted.*

Theorem 2. *At either $\mathrm{A}_1+\mathrm{D}_5+\mathrm{E}_7+\mathrm{E}_8$ or $\mathrm{D}_5+2\mathrm{E}_8$, under the same compact arithmetic and ground-field hypotheses, at most one of the D_5 faces marked $(0,2), (1,3), (1,4)$ is triangular. Every other nonadjacent D_5 face is nontriangular.*

The repository's separate, completed internal compact-dimension–21 field reduction leaves precisely this ground field. That reduction is an input when applying these conditional statements to an arbitrary compact arithmetic chamber in dimension 21. We do not extend the three-field theorem valid from dimension 22 to dimension 21.

2 Actual corners and a complete finite domain

Let T be the positive span of the 21 old vertex roots, and let e be its future unit timelike orthogonal vector. If the face marked a, b is a triangle with third wall c, both root sets replacing a or b by c are positive-definite actual corners.

Lemma 3. *A whole untouched E_8 is orthogonal to c. A whole untouched E_7 is also orthogonal to c.*

Proof. Finite E_8 has no connected finite extension. If c meets a whole E_7, its finite extension must be E_8. Maximality then forces c orthogonal to every other root in either new corner. Their union contains all old roots. The full old forest together with c would therefore be positive definite of rank 22, impossible in signature $(21, 1)$. □

For D_5 marks the remaining complete old domain is $D_5 + A_1$, or just D_5 at the second vertex type. For E_7 or E_8 marks in Theorem 1, we retain respectively the full forests $E_7 + D_5 + A_1$ and $E_8 + D_5 + A_1$. No D_5 or A_1 contact is assumed zero in these latter enumerations.

Lemma 4. *Every contact in these domains has label at most five.*

Proof. A label at least six in a finite Coxeter diagram belongs to an isolated $I_2(m)$. At an old root in a nontrivial connected component this contradicts the retained old neighbors in the two corners: the marks are distinct and nonadjacent. If instead the whole isolated A_1 has such a contact, the isolated-$I_2(m)$ requirement forces c orthogonal to every other old root in both corners, hence their union. Its height then is

$$H = \frac{(2\cos(\pi/m))^2}{2} - 2 < 0,$$

contradicting $c = p + \sqrt{H}e$ with positive height. □

The first finite corner can consequently be enumerated through finite trees. The new root has degree at most three, meets each old connected component at most once, and with degree three has only label-three bonds. Restore the omitted marked root with labels two through five and test the second corner. These conditions contain every actual triangle, not merely a selected finite witness.

Marks	First-corner trials	Second-corner trials	Finite eligible
E_7 in $E_7 + D_5 + A_1$	7497	2296	379
E_8 in $E_8 + D_5 + A_1$	12492	3368	470

For $D_5 + A_1$, a separate direct enumeration checks all $6 \cdot 4^6 = 24576$ contact vectors and has 139 finite-eligible occurrences.

3 Arithmetic and the surviving normals

Put $s = \sqrt{5}$ and $\phi = (1 + s)/2$. The computation uses exact rational arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{5})$, the field containing every contact allowed by Lemma 4. For each old Gram G,

$$H = bG^{-1}b^{\mathsf{T}} - 2.$$

It rejects $H \leq 0$. Simple cycles use c and the unique old path between two contacts; the old path contributes rational factors. Their products must belong to $\mathbb{Q}(s)$. Once that holds, $H \in \mathbb{Q}(s)$, and conjugate positive semidefiniteness requires $\overline{H} \leq 0$. The old rational positive Gram is unchanged, so this follows directly from its Schur complement. Connected affine principal subsets containing c are excluded by compactness: their positive null kernels give a future ideal point in the chamber closure. Finite subsets, and connected subsets with label five, cannot be affine.

The $D_5 + A_1$ domain leaves six roots:

Name	D_5 contacts	A_1 contact	H
C_0	ϕ at0	0	$(s-1)/2$
L_3, L_4	ϕ at3 or4	0	$(-1+5s)/8$
U_3, U_4	1 at1, ϕ at3 or4	0	$(23+13s)/8$
C_*	1 at0	ϕ	$(s-1)/4$

The first and last correspond to marks $(0,2)$; the two fork families correspond to $(1,3)$ and $(1,4)$. In $\mathrm{D}_5 + 2\mathrm{E}_8$ there is no A_1, and only the first five roots occur.

With the whole $\mathrm{D}_5 + \mathrm{A}_1$ included, the exceptional domains leave:

Name	Exceptional contact	A_1 contact	H
E_0	E_7: ϕ at0	0	$1+s$
E_5	E_7: ϕ at5	0	$(1+3s)/4$
E_*	E_7: 1 at5	ϕ	$(1+s)/4$
F_0	E_8: ϕ at0	0	$4+2s$
F_6	E_8: ϕ at6	0	$1+s$

All have zero D_5 contact; this is an output of the complete domains. Their nonadjacent triangle marks are respectively

$$E_0 : (0,2), \quad E_5 : (2,4),(2,5),(3,5), \quad E_* : (3,5),$$
$$F_0 : (0,2), \quad F_6 : (3,5),(3,6),(4,6).$$

4 Coupling at the common vertex

For two candidate walls,

$$c_i \cdot c_j = p_i \cdot p_j - \sqrt{H_i H_j}.$$

The old projections of roots from different components are orthogonal, except for their explicitly shared A_1 contacts. The ring of integers is $\mathbb{Z}[\phi]$: $a + bs$ is integral exactly when $2b$ and $a - b$ are integers. With zero projection product, $H_i H_j$ is the length-two cyclic product and must be integral.

Proof of Theorem 2. The exact $\mathrm{D}_5 + \mathrm{A}_1$ inverse gives positive projection product p and $p^2 - H_i H_j > 0$ for all fifteen pairs of the six candidate walls. Thus each mutual product would be positive, contrary to acuteness. For the new C_* against C_0, L_j, U_j, these positive differences are

$$\frac{3(1+s)}{4}, \qquad \frac{-7+5s}{16}, \qquad \frac{9+5s}{16}.$$

The ten pairs among the old five are independently checked as well. Distinct marked faces use disjoint candidate sets, proving the result. □

Proof of Theorem 1. The fifty-five candidate pairs split into eighteen forbidden positive mutual products, twenty-three nonintegral length-two cycles, two orthogonal necessary pairs, and twelve remaining necessary native or dotted possibilities. The shared A_1 is essential: for C_*, E_* their actual product is $(1+s)/4 > 0$.

Within E_7, E_5 is incompatible with E_0, E_*, whereas E_0, E_* are orthogonal. Thus at most three designated triangles occur there. The two E_8 roots are orthogonal and supply at most four. If any D_5 triangle occurs, every E_7 root except possibly E_0 is forbidden, giving at most $1 + 1 + 4 = 6$. Otherwise the total is at most $3 + 4 = 7$. The exact checker also enumerates all $2^{11} = 2048$ necessary root subsets and gives the unique maximizing subset $\{E_5, F_0, F_6\}$. □

5 Forced exceptional faces and the remaining charge problem

An actual E_8 wall F_0 forces all three triangles on marks $\{0,1,2\}$, and F_6 forces all six on $\{3,4,5,6\}$. The checked positive corners are actual vertices by projection of a strict interior timelike point off a positive Gram span. The inverse is nonnegative, so all other inequalities remain strict. The resulting simplex-edge graphs saturate the degree of each simple face and hence exhaust its connected edge graph. These are actual simplex three- and four-faces.

The inverse entries $J_{00} = 4, J_{66} = 2, J_{06} = 2$ give

$$p_0 \cdot p_6 = 2\phi^2 = 3 + s, \qquad H_0 H_6 = (3 + s)^2,$$

so the walls are orthogonal. If both occur, all twenty corners obtained by omitting one member of $\{0, 1, 2, F_0\}$ and one of $\{3, 4, 5, 6, F_6\}$ are positive. Keeping root7 and the other old components as supports gives a seven-face with the full incidence structure $\Delta^3 \times \Delta^4$, again by degree saturation and connectedness. Its intrinsic dihedral cosine at old roots1,2 is $1/\sqrt{3}$, so it is not intrinsically Coxeter.

For $D_5 + 2E_8$, the earlier complete nonadjacent domain has an upper bound nine on its forty-eight designated triangles. The present D_5 strengthening does not reduce it to eight: one long D_5 triangle and four from each E_8 remain in the necessary domain.

Finally, a nontriangle can be a quadrilateral. The current global rule requires total payment at least two on a triangle and at least one on a quadrilateral. The results here provide no new quadrilateral payment, vertex-allowance saving, dimension–20 theorem, or example. Their next use requires actual quadrilateral or edge compatibility and a verified change to the global charge system.

6 Reproducibility and verification scope

The accompanying proof map is `docs/compact21_supported_triangle_constraints_map.md`. The two complete new domain producers support `--verify`; `check_compact21_e8_triangle_bundles_20261018.py` is a standalone exact arithmetic check of the earlier D_5 and E_8 identities and all twenty product-face corners. Sources and field reduction inputs are hash-bound. Generation and separate full replays pass with measured peak resident memory below 32 MiB. The jobs use one global lock, a 59-second alarm and a periodic one-GiB memory guard. No optimizer or Lean process is launched.

The universal actual-corner, compactness, projection and edge-graph arguments are written proofs. Published finite/affine classifications, compact simplicity and Vinberg's arithmetic criterion are explicit theoretical inputs. The earlier $D_5 + 2E_8$ domain and complete dimension–21 field reduction are separate internally checked inputs. No end-to-end Lean or external human verification is claimed.